

Cross-Sectional Ranking Power Without Alpha:

A Walk-Forward Study of Five Price-Based Equity Factors, 2006–2023

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Abstract

I test whether five standard price-based factors — long-horizon momentum, short-term reversal, realized volatility, volatility-scaled momentum, and medium-horizon momentum — generate market-neutral alpha in the S&P 500 cross-section over 2006–2023. Using a 2.0-million-observation panel, rolling walk-forward validation with purged boundaries, and a beta-neutral long-short portfolio net of transaction costs, I find they do not.

The result is more specific than a simple failure. First, only realized volatility carries standalone predictive power (IC 0.043, $t \approx 2.84$); the three momentum variants are statistically indistinguishable from zero. Second, two of the five factors correlate at 0.91, and the fitted linear model assigns them large opposing coefficients that cancel — deleting three redundant factors nearly doubles out-of-sample information coefficient, from 0.0236 to 0.0448, and raises the share of positive folds from 71% to 86%. Third, and most importantly, the improved signal still loses money. The top predicted decile outperforms the bottom by 1.40% per month, but every decile has a positive mean return, and the spread is fully explained by a long-leg beta of ≈ 1.3 against a short-leg beta of ≈ 0.6 . Imposing beta neutrality forces approximately 30/70 leg weights, which is a structural short of the equity risk premium; the resulting portfolio returns -1.46% annually with a Sharpe ratio of -0.20 .

The paper’s contribution is methodological rather than strategic: it documents a case where cross-sectional ranking power is real, statistically significant, and economically worthless, and it identifies the specific mechanism. A by-product of the design quantifies survivorship bias in the commonly used “current index constituents” universe at roughly 9 percentage points per year.

1 Introduction

Factor investing rests on the claim that observable characteristics predict relative returns. Since fama1993 the literature has catalogued dozens of such characteristics, and mclean2016 document that published anomalies decay substantially after publication — consistent with arbitrage capital competing the return away.

This paper asks a narrow version of the question using only publicly available data and five widely documented signals. The finding is negative, and the value lies in the decomposition: the signal does rank the cross-section, the ranking is statistically significant, and it nevertheless

produces no investable return. Understanding why requires separating three things that are easy to conflate — predictive correlation, portfolio construction, and factor exposure.

Three design decisions are worth stating up front, since each changes the answer materially:

1. **Information coefficient is computed within each date and then averaged**, never pooled. Pooling across a multi-year panel mixes cross-sectional with time-series variation and overstates what a cross-sectional model achieves.
2. **Training and test windows are purged** by the target horizon. Without a 21-day embargo, terminal training observations have targets extending into the test period lopezdeprado2018.
3. **The portfolio is beta-neutral, not dollar-neutral**. As Section 6 shows, this single choice reverses the sign of the strategy's return.

2 Data

2.1 Universe

Constituents are the current S&P 500 membership list (503 tickers), of which seven fail to download because they listed after the sample window. Daily split- and dividend-adjusted closing prices are retrieved from Yahoo Finance for 2005-01-01 through 2024-01-01, yielding 4,781 trading days.

The completeness filter deserves comment. A naive threshold — requiring, say, 95% of available days — would exclude every firm that listed after 2006 and shrink the universe to long-tenured incumbents, worsening the survivorship problem the filter is supposed to be neutral about. Instead I require a three-year minimum history and allow names to enter the panel as they become available. The universe grows from 392 names in 2005 to 488 from 2020 onward, with no discontinuities.

2.2 Survivorship bias

The sample is severely survivor-biased and the bias is not correctable with free data. Membership is the 2026 index applied retrospectively, so the 2005 cross-section consists of 392 firms that were public in 2005 *and* remained in the index two decades later. Delisted, acquired, and demoted firms are absent.

The magnitude is measurable. The universe-average 21-day return is 1.41 1.4118.36 18.369.58 9.588.78 8.78 **percentage points per year**. This overstates pure survivorship bias, since the universe is equal-weighted while SPY is capitalization-weighted, and the small-cap tilt of equal weighting contributes part of the difference. It is therefore an upper bound. All return levels in this paper are inflated by some fraction of it; long-short spreads are less affected, since the bias applies to both legs.

3 Factor construction

Five factors are computed from daily returns and cross-sectionally rank-normalized within each date to $\text{pct_rank} - 0.5$:

Table 1: Factor definitions

Factor	Definition	Window	Reference
<code>mom_12_1</code>	$\sum_{t-252}^{t-21} r$	12m less 1m	jegadeesh1993
<code>reversal</code>	$-\sum_{t-21}^t r$	1m	jegadeesh1990
<code>volatility</code>	$\sigma_{21}(r)$	1m	ang2006
<code>vol_adj_mom</code>	<code>mom_12_1</code> / σ_{21}	12m	risk-scaled trend
<code>mom_3</code>	$\sum_{t-63}^t r$	3m	medium-horizon trend

Rank normalization is a substantive choice, not cosmetic. The models are asked to order stocks against one another on a given date, so market-wide shifts in raw factor levels are noise. Ranking also renders the features scale-free, eliminating the need for a separate standardizer whose parameters would otherwise be fitted across the full sample — itself a subtle leak.

The prediction target is the 21-day forward return, constructed as `rolling(21).sum().shift(-21)`.

The resulting panel contains **2,011,701 stock-day observations** spanning 2006-01-03 to 2023-11-29, with a median of 454 names per date. The 252-day momentum window consumes 2005; the forward target truncates the final 21 days.

4 Methodology

4.1 Walk-forward validation

A single chronological split evaluates one regime. I instead train on 756 trading days (≈ 3 years), skip 21 days, and test on the following 252 days, advancing one year per iteration. This produces **14 folds with non-overlapping test windows**, February 2009 through February 2023.

The 21-day skip implements purging in the sense of lopezdeprado2018. Because the target spans 21 days, the final training observations would otherwise carry information from the test period.

A limitation: the first training block consumes 2006–2008, so the earliest test window opens after the financial crisis. The models are evaluated on the recovery, never on the crisis itself.

4.2 Models

Ridge regression ($\alpha = 1.0$) serves as the linear specification. Gradient boosting (XGBoost; 300 trees, depth 4, learning rate 0.05, subsample 0.8, column-subsample 0.8) provides the nonlinear comparison. Hyperparameters were fixed a priori rather than tuned, to avoid consuming the out-of-sample data through selection.

4.3 Evaluation

The information coefficient is the Spearman rank correlation between predicted and realized forward returns, computed *within each date* and then averaged:

$$\overline{\text{IC}} = \frac{1}{T} \sum_{t=1}^T \rho_{\text{Spearman}}(\hat{y}_t, y_t).$$

For a return series, the t -statistic on annualized alpha is approximately $\text{SR}\sqrt{\text{years}}$. At a Sharpe ratio of 0.3, roughly 44 years are required to reach $t = 2$ — a constraint that bounds what any study of this length can establish and which motivates the 18-year sample.

5 Results

5.1 Walk-forward performance

Table 2: Walk-forward IC, five factors, 14 folds

Model	Mean IC	Std	Folds > 0	t
Ridge	+0.0236	0.0589	71%	1.50
XGBoost	+0.0121	0.0212	79%	2.14

Ridge posts the higher mean but is markedly less stable, and its average is dominated by one fold: 2009 returns +0.176, seven times the overall mean. Excluding it, Ridge’s average falls to approximately +0.011. The 2016 fold is the mirror image at -0.085 . Both are the same mechanism — the factor set loads on high-volatility names, which rebound sharply in a recovery and decay in a calm market.

XGBoost exhibits neither extreme. Its fold-level information ratio is 0.57 against Ridge’s 0.40, and it clears $t = 2$ where Ridge does not.

This has a practical implication beyond the specific models. Ridge’s higher mean is not evidence of a better model. A single fold contributes more than half of its average, and its fold-level t -statistic (1.50 1.50) fails to clear conventional significance where XGBoost’s (2.14 2.14) does. Stability across regimes, not mean IC, distinguishes the two.

5.2 Factor diagnostics

Table 3: Standalone IC by factor (t adjusted for 21-day overlap)

Factor	Mean IC	Std	t
volatility	+0.0429	0.2212	+2.84
reversal	+0.0188	0.1565	+1.76
mom_12_1	+0.0006	0.1900	+0.05
vol_adj_mom	−0.0042	0.1817	−0.34
mom_3	−0.0130	0.1738	−1.09

Only realized volatility attains conventional significance. Reversal is marginal. All three momentum specifications are zero or negative over this sample — consistent with mclean2016 on post-publication decay, given that momentum has been documented and traded since 1993.

Table 4: Factor correlation matrix

	mom_12_1	reversal	volatility	vol_adj_mom	mom_3
mom_12_1	1.00	0.00	0.05	0.91	0.32
reversal	0.00	1.00	−0.01	−0.00	−0.53
volatility	0.05	−0.01	1.00	−0.21	−0.01
vol_adj_mom	0.91	−0.00	−0.21	1.00	0.31
mom_3	0.32	−0.53	−0.01	0.31	1.00

mom_12_1 and vol_adj_mom correlate at 0.91: dividing by volatility barely perturbs the cross-sectional ordering, so the two are effectively one variable entered twice. mom_3 correlates −0.53 with reversal, both being constructed from overlapping recent-return windows of opposite sign. Five nominal factors reduce to approximately two independent signals.

The fitted coefficients display the consequence directly. In the terminal fold Ridge assigns −0.0305 to mom_12_1 and +0.0280 to vol_adj_mom — two inputs correlated at 0.91 receiving large opposing weights that nearly cancel. The model is fitting the residual difference between near-identical variables, which is close to pure noise. This is the mechanical source of the fold-to-fold instability, and it explains the 2009 outlier more parsimoniously than any regime narrative.

5.3 The effect of removing redundant factors

Table 5: Walk-forward IC: five factors versus two

Specification	Mean IC	Std	Folds > 0	t
5-factor Ridge	+0.0236	0.0589	71%	≈ 1.50
2-factor Ridge (volatility, reversal)	+0.0448	0.0513	86%	≈ 3.26

Discarding three of five factors nearly doubles mean out-of-sample IC, reduces its dispersion, and raises the positive-fold rate by fifteen percentage points. The multicollinearity diagnosis is thus confirmed out-of-sample: the removed factors were not contributing information, they were destabilizing the fit.

The t -statistic treats the fourteen fold means as independent. Test windows do not overlap, but training windows do, so it should be read as directional.

6 Portfolio results and the source of the spread

6.1 Construction

At each month-end the model’s predictions are ranked cross-sectionally. The top quintile is held long and the bottom quintile short until the next rebalance. Positions are lagged one day, so the portfolio trades on the prior day’s signal.

Leg weights impose *beta* neutrality rather than dollar neutrality. With β_L and β_S the mean trailing 252-day betas of the two legs,

$$w_L = \frac{\beta_S}{\beta_L + \beta_S}, \quad w_S = \frac{\beta_L}{\beta_L + \beta_S},$$

which sets net beta to zero at unit gross exposure. Transaction costs of 16 bps (5 spread, 10 impact, 1 commission) are applied to one-way turnover.

6.2 Performance

Table 6: Beta-neutral long-short performance, net of costs, 2009–2023

	5-factor (XGBoost)	2-factor (Ridge)
Annualized return	−1.54%	−1.46%
Annualized volatility	4.52%	6.41%
Sharpe ratio (net)	−0.321	−0.197
t -statistic	−1.20	−0.74
Maximum drawdown	−36.18%	−33.25%
Residual beta	0.009	0.007
Annualized alpha	−1.59%	−1.37%

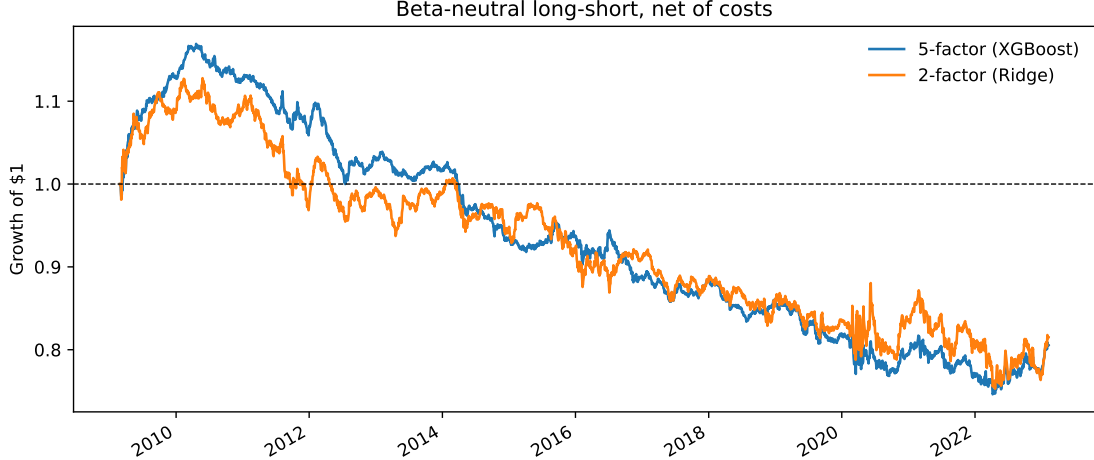


Figure 1: Beta-neutral long-short portfolios, net of 16bps costs. Both specifications decline steadily from 2010 despite positive walk-forward IC.

Beta neutralization is effective — residual beta is under 0.01 with $R^2 \approx 0$. Both specifications lose money. The improvement in IC from dropping redundant factors does not translate into return.

The drawdown profile is a twelve-year grind (peak 2010-04, trough 2022-04) rather than a shock; 48.1% of days are positive; excess kurtosis of 7.25 with the worst days clustered in March–April 2020 reflects gap risk in a long-short book during a volatility spike. Inverting the signal produces a Sharpe ratio of 0.321 with $t = 1.20$, also insignificant — so the strategy is noise centred slightly below zero, not a reliably incorrect model that could be traded in reverse.

6.3 Decile decomposition

Table 7: Mean 21-day forward return by predicted decile, 2-factor model

0	1	2	3	4	5	6	7	8	9
1.194	1.422	1.625	1.689	1.680	1.665	1.695	1.847	2.115	2.590

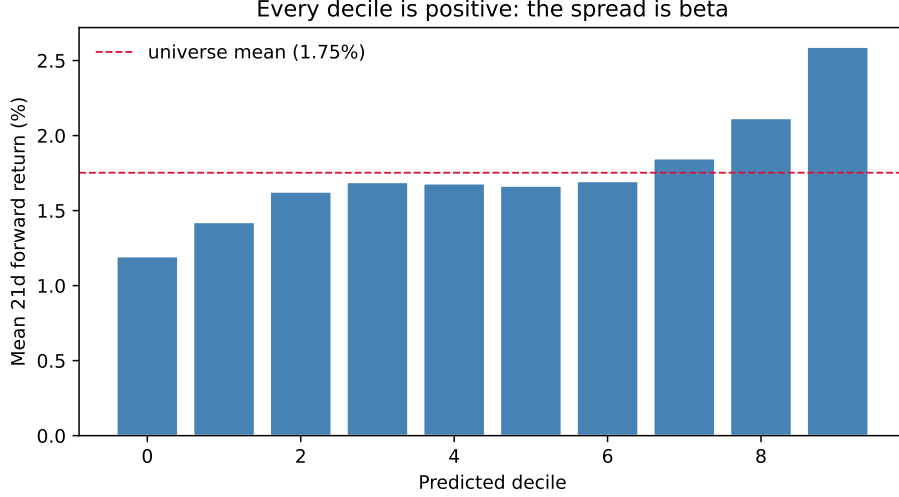


Figure 2: Mean 21-day forward return by predicted decile. The ranking is near-monotonic, but every decile is positive — the spread reflects a beta differential rather than alpha.

The spread from bottom to top decile is +1.396% per 21 days, approximately monotonic outside a flat region in deciles 3–6. The signal ranks the cross-section and the ranking holds in the tails, ruling out the hypothesis that it predicts only the interior of the distribution.

Every decile has a positive mean return. That is what defeats the strategy. Long-leg beta is approximately 1.3 against a short-leg beta of approximately 0.6, so beta neutrality forces weights near 30/70. Applying those to the traded quintiles:

$$\underbrace{0.30 \times 2.353\%}_{\text{long}} - \underbrace{0.70 \times 1.308\%}_{\text{short}} = 0.706\% - 0.916\% = -0.210\% \text{ per 21 days} \approx -2.5\%/yr,$$

which reproduces the realized -1.46% to within costs and time variation in the betas.

The decile spread is compensation for market beta, not alpha. Imposing beta neutrality converts a levered short position in low-beta stocks into a structural short of the equity risk premium, and that carry is negative.

This is the betting-against-beta relationship of frazzini2014 operating in reverse. BAB is long low-beta levered and short high-beta; a volatility-ranking signal does the opposite, so its beta-neutral form is negative by construction. Its dollar-neutral form earns a positive return, but that return is market exposure rather than alpha.

7 Conclusion

Five widely documented price-based factors do not produce market-neutral alpha in the large-capitalization US cross-section over 2006–2023. The specific findings:

1. Only realized volatility carries standalone predictive power (IC 0.043, $t \approx 2.84$). The three

momentum variants are indistinguishable from zero.

2. Redundant factors are harmful, not merely useless. Two of five correlate at 0.91 and receive cancelling coefficients; removing three factors nearly doubles out-of-sample IC.
3. Mean IC alone is a poor model-selection criterion. Ridge posts the higher average but fails significance and depends on one fold; XGBoost is lower but stable and significant.
4. Cross-sectional ranking power does not imply alpha. A 1.40% monthly decile spread is fully accounted for by a beta differential, and vanishes under beta neutrality.
5. Neither specification is investable, and neither is the inverted signal.

The methodological point generalizes beyond these five factors. A researcher evaluating only pooled IC, on a short sample, with a dollar-neutral portfolio, would have concluded that this factor set works. Each of those three choices is individually defensible and collectively produces a false positive.

Reproducibility

Code and data-construction scripts: <https://github.com/yamare1/stat-arb-factor-model>. All results derive from freely available Yahoo Finance data and execute end to end from a cold kernel.

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